

SIOC 251: Radiative Transfer

Homework 4: Two-Stream SW Radiative Transfer and the Aerosol Direct Radiative Effect

1 Overview

The atmosphere over the Arabian Sea contains both mineral aerosols advected primarily southward from the deserts of Afghanistan and Iran, and pollution aerosols emitted from South Asia.

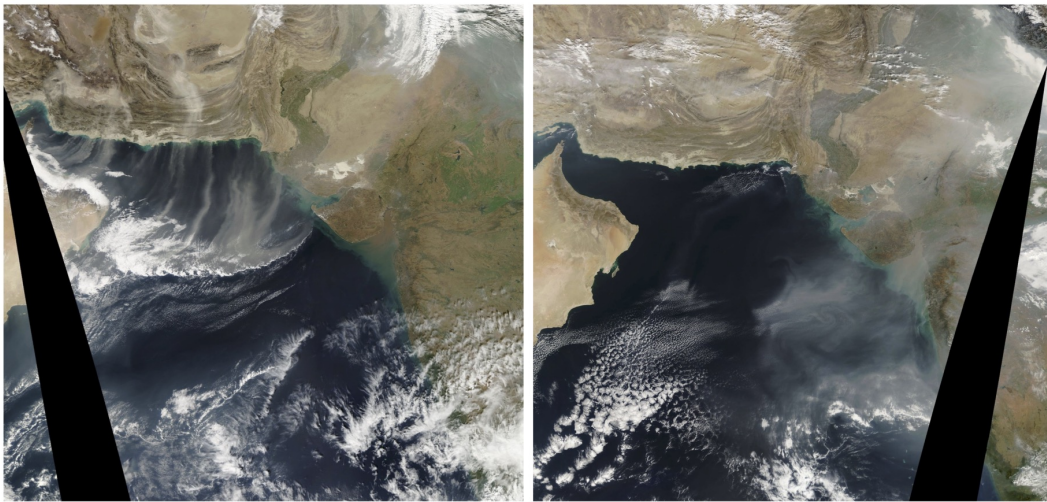


Figure 1: MODIS RGB imagery of dust (left) and pollution (right) advected over the Arabian Sea.

In this homework you will calculate the top-of-the-atmosphere (TOA), surface, and atmospheric direct radiative effect of these different aerosol species in the solar part of the spectrum. The goals of this assignment are to learn to use the two-stream solution to the radiative transfer equation, understand how aerosol single scatter properties vary as a function of wavelength, and explore how different aerosol species affect Earth's climate. You will:

- construct a spectrally resolved shortwave two-stream radiative transfer model
- include aerosol and Rayleigh scattering layers
- calculate spectral and broadband shortwave fluxes
- estimate the TOA, surface, and atmospheric aerosol direct radiative effect
- compare the radiative effects of dust and pollution aerosols

For optional background reading, see: Ramanathan et al., *Aerosols, Climate, and the Hydrological Cycle*, Science, 294, 2119–2124, 2001.

2 Repository Structure and Submission

```
sioc251_radiative_transfer
|
|-- HW1_monte_carlo_extinction
|
|-- HW2_longwave_model
|
|-- HW3_monte_carlo_scattering
|
|-- HW4_two_stream_aerosol_DRE
    |
    |-- two_stream_aerosol_DRE.ipynb
    |
    |-- data
    |   |-- aerosol_properties.csv
    |   |-- ocean_reflectance.csv
    |   |-- Fo.csv
    |
    |-- figures
        |-- figure1.png
```

- Final submission must be a **Python notebook**
- Save figures in **figures/**
- Push to your GitHub repository

3 AI Use Policy

Students are encouraged to use AI tools (e.g., ChatGPT, Claude, Gemini) to assist with coding, debugging, plotting, and documentation. However, students remain fully responsible for the scientific correctness and physical interpretation of their work. AI output should be treated like information from other sources: it must be understood, verified, and appropriately documented. At the end of your notebook, include a short statement describing:

- which AI tools you used, if any
- how you used them
- what parts of the work you checked independently
- at least one example of something you verified rather than simply accepting from AI output

4 Definitions

Let's define the net shortwave radiative flux with positive in the downward direction

$$F^N = F^\downarrow - F^\uparrow, \quad (1)$$

where $F^\downarrow$ is the downward flux and $F^\uparrow$ is the upward flux. The direct radiative effect is

$$\text{DRE} = F^N - F_p^N, \quad (2)$$

where F_p^N is the pristine-sky, or aerosol-free, net radiative flux.

At the top of the atmosphere

$$\text{DRE}_{\text{TOA}} = F_p^\uparrow - F^\uparrow, \quad (3)$$

since the incoming solar flux is independent of the atmospheric state. At the surface we can assume that the upward shortwave flux is related to the downward shortwave flux by

$$F_s^\uparrow = r_s F_s^\downarrow, \quad (4)$$

where r_s is the surface reflectance, so that

$$\text{DRE}_{\text{sfc}} = (F^\downarrow - F_p^\downarrow) (1 - r_s) \quad (5)$$

The TOA, surface, and atmospheric net fluxes must satisfy

$$F_{\text{TOA}}^N = F_{\text{sfc}}^N + F_{\text{atm}}^N \quad (6)$$

Therefore, the atmospheric direct radiative effect can be calculated as

$$\text{DRE}_{\text{atm}} = \text{DRE}_{\text{TOA}} - \text{DRE}_{\text{sfc}} \quad (7)$$

5 Two-Stream Model Construction

Using your class notes or the textbook, create a two-stream SW radiative transfer model you can use to estimate the aerosol direct radiative effect.

5.1 Model Specifics

- a. Define the spectral resolution of your model to be

$$\Delta\lambda = 0.1 \text{ } \mu\text{m}, \quad (8)$$

with wavelength λ ranging from 0.4 to 4 μm .

- b. Read in the spectrally resolved aerosol optical properties from `aerosol_properties.csv`. These include the extinction coefficient normalized to one at $\lambda = 0.55 \text{ } \mu\text{m}$, the single scatter albedo $\tilde{\omega}$, and the asymmetry parameter g for both dust and pollution aerosols.
- c. Read in the spectral ocean surface reflectance from `ocean_reflectance.csv`.
- d. Incorporate a Rayleigh scattering layer into your atmosphere. Assume that the Rayleigh optical depth at $\lambda = 0.55 \text{ } \mu\text{m}$ is

$$\tau_{R,0.55} = 0.1, \quad (9)$$

and that the Rayleigh layer is below the aerosol layer.

- e. For the aerosol layer, use an aerosol optical depth at 0.55 μm of

$$\tau_{a,0.55} = 1.0. \quad (10)$$

5.2 Verification

Demonstrate that your model behaves correctly via two model verification tests that examine model behavior for limiting cases.

5.3 Question

For each aerosol type, calculate the spectrally resolved transmittance (at the surface) and albedo (at the TOA). Are these variables and their differences between the two aerosol layers consistent with the aerosol single scatter properties?

6 Part II: Shortwave Fluxes and the DRE

We will next use the model to estimate the DRE at the surface, TOA, and in the atmosphere, for both aerosol species.

6.1 Model Construction

- Read in the spectrally resolved top-of-atmosphere downwelling solar flux, $F_0(\lambda)$, from `Fo.csv`.
- Use your model output to calculate the spectrally resolved upward and downward shortwave fluxes at the top of the atmosphere and surface, and then integrate to estimate the SW fluxes.
- Use these data to calculate the TOA, surface, and atmospheric instantaneous DRE for the dust and pollution aerosols.

6.2 Verification

Add one test to convince yourself that your fluxes and DRE values are reasonable.

6.3 Questions

Use your model to answer the following questions

- For $\tau_{a,0.55} = 1.0$, what is the instantaneous DRE at the top of the atmosphere for dust and pollution aerosols and what is the cause of any differences?
- Based on the sign of the TOA DRE, do these aerosols warm or cool the climate system?
- How do you think the presence of these aerosols may influence convection?
- Do changes in surface reflectance have a similar effect on both types of aerosol species?
- The forcing efficiency FE is the change in the DRE per unit change in $\tau_{a,0.55}$

$$\text{FE} = \frac{d\text{DRE}}{d\tau_{a,0.55}}$$

Which aerosol species has a larger TOA forcing efficiency and why?

7 Part III: Student Investigation

Come up with your own question & hypothesis that can be answered by the model, run the simulations, interpret the results, and answer your question.